

Given: Due:	Further Maths (Core Pure) HW 03	Name:
Chapter 2: Complex Numbers: loci		
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.		
91% (A*)	76% (A)	66% (B)
56% (C)	50% (D)	40% (E)
Useful resources: <i>Useful resources:</i> Your notes from recent lessons & your text book <u>Although most Complex arithmetic can be done on your calculator, show all steps to reach the answer & use calculator for checking results only.</u>		
Questions - do these on lined paper		
(a) Shade on an Argand diagram the set of points $\left\{ z \in \mathbb{C} : z - 4i \leq 3 \right\} \cap \left\{ z \in \mathbb{C} : -\frac{\pi}{2} < \arg(z + 3 - 4i) \leq \frac{\pi}{4} \right\}$ (6) The complex number w satisfies $ w - 4i = 3$ (b) Find the maximum value of $\arg w$ in the interval $(-\pi, \pi]$. Give your answer in radians correct to 2 decimal places. (2)		
(a) Shade on an Argand diagram the set of points $\left\{ z \in \mathbb{C} : z - 1 - i \leq 3 \right\} \cap \left\{ z \in \mathbb{C} : \frac{\pi}{4} \leq \arg(z - 2) \leq \frac{3\pi}{4} \right\}$ (5) The complex number w satisfies $ w - 1 - i = 3 \text{ and } \arg(w - 2) = \frac{\pi}{4}$ (b) Find, in simplest form, the exact value of $w ^2$ (4)		
$z - 1 - i = 4, z \in \mathbb{C}.$ a) Sketch the locus of the points that satisfy the above equation in a standard Argand diagram. b) Find the minimum and maximum values of z for points that lie on this locus. (6)		

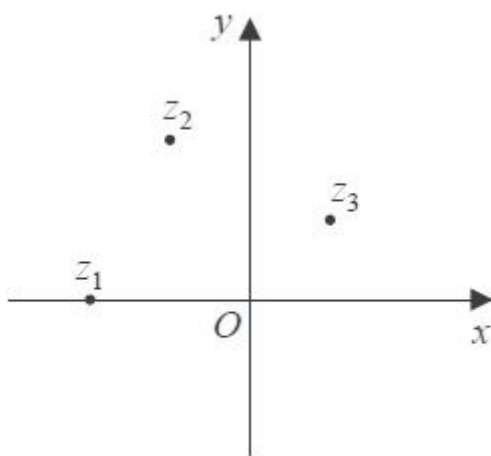


Figure 1

The complex numbers $z_1 = -2$, $z_2 = -1 + 2i$ and $z_3 = 1 + i$ are plotted in Figure 1, on an Argand diagram for the complex plane with $z = x + iy$

(a) Explain why z_1 , z_2 and z_3 cannot all be roots of a quartic polynomial equation with real coefficients.

(2)

(b) Show that $\arg \left(\frac{z_2 - z_1}{z_3 - z_1} \right) = \frac{\pi}{4}$

(3)

(c) Hence show that $\arctan(2) - \arctan\left(\frac{1}{3}\right) = \frac{\pi}{4}$

(2)

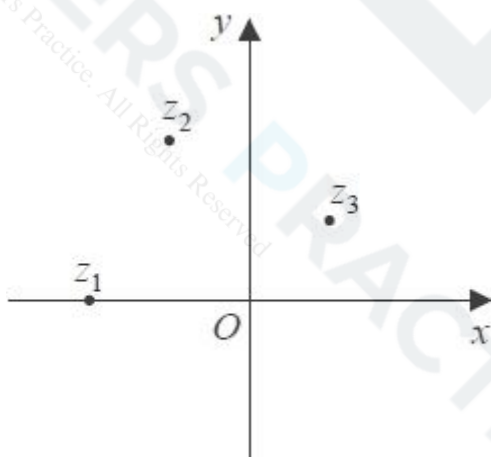


Diagram 1

(d) Shade, on Diagram 1, the set of points of the complex plane that satisfy the inequality

$$|z + 2| \leq |z - 1 - i|$$

(2)

END OF HOMEWORK

Total = 32 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.